

Homework 3

《生物医学图像处理》

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1 Random noise in MR Images follows a Rician distribution in magnitude images:

$$p(M|I, \sigma) = \frac{M}{\sigma^2} e^{-\frac{M^2+I^2}{2\sigma^2}} I_0\left(\frac{MI}{\sigma^2}\right)$$

Where I is the true underlying signal intensity, M is the noisy measured magnitude, σ^2 is the noise variance in the complex K-space domain, and I_0 is the 0-th order modified Bessel function.

$$I_0(0) = 1, \quad I_0(z) \approx \frac{e^z}{\sqrt{2\pi z}}, \quad z \gg 1$$

- (1) Show the derivation when the true signal $I = 0$. State its final Probability Density Function (PDF), to prove Rician is similar to Rayleigh in the background.
- (2) Derive the PDF when the local Signal-to-Noise Ratio (SNR) is high ($I \gg \sigma$) to prove that it approaches a normal distribution $N(I, \sigma^2)$. (Hint: Make good use of approximations)
- (3) Based on (1) and (2), if you were to design an adaptive Alpha-Trimmed Mean filter, how would you set the value of d for the background area and the tissue area? Just provide a simple qualitative explanation. (In engineering practice, MRI reconstruction often employs non-local filtering. However, the objective of this question is to evaluate your fundamental understanding of the relationship between noise distributions and adaptive filtering strategies.)

SOLUTION:

(1) 当 $I = 0$ 时,

$$p(M|I, \sigma) = \frac{M}{\sigma^2} e^{-\frac{M^2+I^2}{2\sigma^2}} I_0(0) = \frac{M}{\sigma^2} e^{-\frac{M^2}{2\sigma^2}}$$

已知瑞利分布在背景中的概率密度函数为:

$$p(z) = \frac{2z}{b} e^{-\frac{z^2}{b}} \quad (z \geq 0)$$

令 $z = M$, $b = 2\sigma^2$, 可知两式等价。因此证明了在背景区域, 其噪声服从瑞利分布。

(1) 当 $I \gg \sigma$ 时, $\frac{MI}{\sigma^2} \gg 1$, 因此

$$p(M|I, \sigma) = \frac{M}{\sigma^2} e^{-\frac{M^2+I^2}{2\sigma^2}} \frac{e^{\frac{MI}{\sigma^2}}}{\sqrt{2\pi \frac{MI}{\sigma^2}}} = \frac{1}{\sigma\sqrt{2\pi}} \sqrt{\frac{M}{I}} e^{-\frac{M^2-2MI+I^2}{2\sigma^2}} = \frac{1}{\sigma\sqrt{2\pi}} \sqrt{\frac{M}{I}} e^{-\frac{(M-I)^2}{2\sigma^2}}$$

由于 $I \gg \sigma$, 因此 $M \approx I$, 所以

$$p(M|I, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(M-I)^2}{2\sigma^2}} \sim N(I, \sigma^2)$$

(3) 对于背景区域：应该设置较大的 d 值。

由第一问可知，背景区域服从瑞利分布。瑞利分布是一种明显的偏态分布（右偏/长尾），会产生一些孤立的、较亮的假性噪声点。较大的 d 值能让滤波器更接近中值滤波，从而有效剔除这些位于分布尾部的极端噪声值，防止背景被均值拉亮。

对于组织区域：应该设置较小的 d 值。

由第二问可知，组织区域信噪比较高，噪声服从对称的正态分布。对于高斯噪声，均值滤波（ $d = 0$ ）是数学上最优的线性无偏估计。因此，使用较小的 d 可以在有效平滑高斯噪声的同时，最大限度地保留组织的解剖学细节。

2 Constrained Least Squares Filtering:

$$\hat{F}(u, v) = \left[\frac{H^*(u, v)}{|H(u, v)|^2 + \gamma |P(u, v)|^2} \right] G(u, v), \quad p(x, y) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

- (1) Prove that $P(u, v) = 4 - 2 \cos\left(\frac{2\pi u}{M}\right) - 2 \cos\left(\frac{2\pi v}{N}\right)$
- (2) Discuss $\gamma = 0$, what does the CLS filter become? $\gamma \rightarrow +\infty$, what happens to the restored image?
- (3) Consider an image degraded by uniform linear motion.

$$H(u, v) = \frac{T}{\pi(ua + vb)} \sin[\pi(ua + vb)] e^{-j\pi(ua + vb)}$$

Assume there is slight additive noise $\eta(x, y)$ in the motion-blurred image. When we use CLS Filtering, explain why setting $\gamma = 10^{-6}$ may be significantly better than $\gamma = 0$. Just provide a simple qualitative explanation.

SOLUTION:

$$\begin{aligned} (1) \quad P(u, v) &= \sum_x^M \sum_y^N p(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \\ &= 4 - (e^{-j2\pi\frac{v}{N}} + e^{j2\pi\frac{v}{N}}) - (e^{-j2\pi\frac{u}{M}} + e^{j2\pi\frac{u}{M}}) \\ &= 4 - 2 \cos\left(\frac{2\pi u}{M}\right) - 2 \cos\left(\frac{2\pi v}{N}\right) \end{aligned}$$

(2) 当 $\gamma = 0$ 时,

$$\begin{aligned} \hat{F}(u, v) &= \frac{H^*(u, v)}{|H(u, v)|^2} G(u, v) \\ &= \frac{H^*(u, v)}{H(u, v)H^*(u, v)} G(u, v) \\ &= \frac{G(u, v)}{H(u, v)} \end{aligned}$$

CLS 滤波器退化为了直接逆滤波器。

当 $\gamma \rightarrow +\infty$ 时:

对于 $P(u, v) \neq 0$ 的高频区域, 分母中的 $\gamma |P(u, v)|^2$ 项趋于无穷大, 导致滤波器增益趋近于 0, 图像细节被抹除。恢复出的图像会丢失所有频率信息, 图像的像素值趋于 0 (变成全黑), 或者如果是计算中保留了直流分量, 图像将变成没有任何细节和边缘的、极其平滑的纯色均匀图像。

由于 $P(u, v)$ 在原点处 $P(0, 0) = 4 - 2 \cos(0) - 2 \cos(0) = 0$, 在该频率点 (直流分量):

$$\hat{F}(0, 0) = \frac{H^*(0, 0)}{|H(0, 0)|^2 + \gamma \cdot 0} G(0, 0) = \frac{G(0, 0)}{H(0, 0)}$$

(1) 退化函数 $H(u, v) = \frac{T}{\pi(ua+vb)} \sin[\pi(ua+vb)]e^{-j\pi(ua+vb)}$ ，这个函数具有周期性的过零点（即在特定的频率上， $\sin[\pi(ua+vb)] = 0$ ，导致 $H(u, v) = 0$ ）。

如果设置 $\gamma = 0$ （即使用直接逆滤波），重建公式包含除以 $H(u, v)$ 。在 $H(u, v)$ 接近 0 的频率点，轻微的噪声 $N(u, v)$ 会被极大地放大（即 $\frac{N(u, v)}{H(u, v)} \rightarrow \infty$ ），这会在图像上放大噪声，完全掩盖真实的图像信息。

如果设置 $\gamma = 10^{-6}$ ，在 $H(u, v) \rightarrow 0$ 的过零点处，分母 $|H(u, v)|^2 + \gamma|P(u, v)|^2$ 不再为零，而是由 $\gamma|P(u, v)|^2$ 决定值。拉普拉斯算子 $|P(u, v)|^2$ 在高频处是一个较大的正值，避免了分母变为 0，压制了高频噪声的无限制放大。

因此， $\gamma = 10^{-6}$ 能在有效去除运动模糊的同时控制住噪声，远好于 $\gamma = 0$ 。

3 Suppose an image's degradation function is the convolution of original image and $h(x, y)$ and here assume x, y are continuous.

$$h(x, y) = \frac{x^2 + y^2 - 2\sigma^2}{\sigma^4} e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

- (1) Prove that the degradation in frequency filed is $H(u, v) = -8\pi^3\sigma^2(u^2 + v^2)e^{-2\pi^2\sigma^2(u^2 + v^2)}$
- (2) Assume the power spectrum ratio of noise and undegraded image $\left(\frac{S_n}{S_f}\right)$ is the constant parameter K , give the transfer function expression of Wiener filter for this image.

SOLUTION:

$$(1) \quad h(x, y) = \left(\frac{x^2 - \sigma^2}{\sigma^4} + \frac{y^2 - \sigma^2}{\sigma^4} \right) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

考虑 $f(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}}$ 为高斯函数，高斯函数的二维傅里叶变换仍为高斯函数，查公式得：

$$F(u, v) = 2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2 + v^2)}$$

令 $f(x, y)$ 对 x 和 y 求二阶偏导

$$\frac{\partial f}{\partial x} = -\frac{x}{\sigma^2} e^{-\frac{x^2 + y^2}{2\sigma^2}}, \quad \frac{\partial^2 f}{\partial x^2} = \frac{x^2 - \sigma^2}{\sigma^4} e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

$$\frac{\partial f}{\partial y} = -\frac{y}{\sigma^2} e^{-\frac{x^2 + y^2}{2\sigma^2}}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{y^2 - \sigma^2}{\sigma^4} e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

$$h(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

因此，

$$\begin{aligned} H(u, v) &= [(j2\pi u)^2 + (j2\pi v)^2] F(u, v) \\ &= -8\pi^3\sigma^2(u^2 + v^2)e^{-2\pi^2\sigma^2(u^2 + v^2)} \end{aligned}$$

(2) 第一题所求的 $H(u, v)$ 为纯实数函数，因此维纳函数

$$\begin{aligned} W(u, v) &= \frac{1}{H(u, v)} \frac{H^2(u, v)}{H^2(u, v) + \frac{S_n}{S_f}} \\ &= -\frac{8\pi^3\sigma^2(u^2 + v^2)e^{-2\pi^2\sigma^2(u^2 + v^2)}}{64\pi^6\sigma^4(u^2 + v^2)^2 e^{-4\pi^2\sigma^2(u^2 + v^2)} + K} \end{aligned}$$